

Past paper (Long)

- Q. What is recovery? write steps for recovering instance, data file and control files?
(Repeat 2-times)

Recovery

⇒ Refers to the process of restoring the database to a consistent state after a failure or data loss.

⇒ Restoring can involve restoring backups.

Instance Recovery

⇒ Instance recovery is typically automatic and is required when an Oracle instance fails unexpectedly.

The database performs instance recovery automatically at startup if needed.

⇒ The steps involved are:

1. Crash Detection:

Oracle detects that an instance has crashed.

2. Redo Application:

During the startup Oracle applies the redo logs to ensure all committed transactions are applied to the data files.

3. Rollback uncommitted transactions:

Rollback any uncommitted transactions to ensure data consistency.

2. Datafile Recovery

⇒ Datafile recovery is necessary when one or more data files are lost or corrupted. The steps to recover data files are as follows:

1. Identify the affected data files.

Determine which data files need recovery.

2. Take the tablespace offline (optional):

ALTER TABLESPACE tablespace name

OFFLINE IMMEDIATE;

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3. Restore the data files from backup.
⇒ Using RMAN:

RMAN > RESTORE DATAFILE 'datafile-path'

⇒ if using OS utility, copy the datafile from backup location to the original location.

4. Recovering the data files:

Using RMAN:

RMAN > RECOVER DATAFILE 'datafile-path';

5. Bring the tablespace online. Path:

Using SQL*PLUS:

RECOVER DATAFILE 'datafile-path';

Bring the tablespace online:

ALTER TABLESPACE tablespace-name
ONLINE;

3. Control file recovery

Require some or all control files are corrupted.

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1. Identify the last control files:
Check the alert.log or trace files for errors related to control files.

2. Restore the control files from backup.

Using RMAN

RMAN > RESTORE CONTROLFILE

Use OS for backup to original location

3. Mount the DB start:

⇒ STARTUP NOMOUNT;

RMAN > RESTORE CONTROLFILE;

5. Mount the DB: Restore the control files.

• ALTER DATABASE MOUNT;

6. Recover the DB:

- RMAN > RECOVER DATABASE
- RECOVER DATABASE USING BACKUP CONTROLFILE;

Open the DB:

- ALTER DATABASE OPEN RESETLOGS;

Steps for recovering
redo log files and control
files? (Repeat 2 times)

1. Recovering redo logs in ARCHIVELOG
mode:

Identifying the last redo log files:
⇒ Check the alert and
trace file for checking the
corrupted log files.

2. Take the affected redo log group
offline:

⇒ ALTER DATABASE CLEAR UNARCHIVED
LOGFILE GROUP grp.num;

3. Drop the redo log group if
necessary:

⇒ ALTER DATABASE DROP LOGFILE
GROUP grp.num;

4. ADD a new redo log group.

```
ALTER DATABASE
```

```
ADD LOGFILE (GROUP grp_num ('path to new file'))
```

```
SIZE size-in-MB;
```

5. Verify the new redo log group.

```
SELECT GROUP#
```

```
STATUS FROM V$LOG;
```

Redo log file loss in Non-Archive mode

1. Shutdown the database immediately.

```
⇒ SHUTDOWN IMMEDIATE;
```

2. Restore the entire database from backup.

```
⇒ Restore all data files, control files & redo log files from the recent backup.
```

3. Start the DB in mount mode.

```
⇒ START MOUNT;
```

4. Perform media recovery.

```
⇒ RECOVER DATABASE;
```

5. Open the DB.

```
⇒ ALTER DATABASE OPEN RESETLOGS;
```

Q. Write the procedure to resize smallfile Tablespace using ALTER DATABASE?

Check the current tablespace size.

```
SELECT Filename, Bytes, maxBytes, Autoextend
```

```
FROM DBA_DATA_FILES
```

```
WHERE Tablespace-name = 'name';
```

Determine the desired size.

Alter the TS size;

```
⇒ ALTER DATABASE
```

```
DATAFILE 'full path to -datafile'
```

```
RESIZE 10GB;
```

Verify the change

Q. Write note on how undo tablespaces are managed by creating, dropping and modifying undo tablespaces?

1) Create undo tablespace:

```
CREATE UNDO TABLESPACE undotbs1
DATAFILE '/u01/path/... .dbf'
SIZE 500M
AUTOEXTEND ON
NEXT 100M
MAXSIZE UNLIMITED;
```

2) Dropping an undo tablespace:

↳ enabling or disabling undo retention guarantee.

```
DROP TABLESPACE undotbs1
INCLUDING CONTENTS AND DATAFILES;
```

3) Switching undo tablespace:

```
ALTER SYSTEM SET UNDO_TABLESPACE=undotbs1;
```

Adjust undo retention:

```
ALTER SYSTEM SET
```

```
UNDO_RETENTION = 900; -- in second retention period.
```

Resize undo tablespace:

```
ALTER DATABASE
```

```
DATAFILE 'path.dbf'
```

```
RESIZE 1G;
```

Auto extend configuration:

```
ALTER DATABASE
```

```
DATAFILE 'path1.dbf'
```

```
AUTOEXTEND ON
```

```
NEXT 100M
```

```
MAXSIZE 10G;
```


Q- Explain the role and functioning of Oracle instance?

Oracle server: The power house for enterprise data.
[elaboration on register 1]

Functionalities of Oracle server:

1- Data Management:

DDL: Creates, modifies and deletes database objects like tables, indexes & views.

DML: Inserts, updates and deletes data within database.

DCL: Grants, revokes and manages user access privileges to ensure data security.

2- Query processing:

SQL: Enable users to retrieve, manipulate and analyze data through powerful queries

indexes:
views:

3- Transaction management:

ACID property: Ensure data consistency and integrity through ACID property

Rollback:

4- Security:

- User Authentication
- Data encryption
- Access Control

5- Backup & Recovery:

- RMAN
- Data Guard

6- Administration:

- SQL*Plus
- OEM

Oracle instance:

The heart of database operations

Functionalities:

- Background processes
- Memory structures
- Datafiles management

Note:

All solved on registers 1.

Q. What is ROWID Pseudo column?
ROWID Pseudo columns return the address of each row in the table. Oracle assigns a ROWID to each row.

ROWID consists of:

The data object number of the object.

The datablock is the datafile in which the row resides.

The position of the row in the data block.

The datafile in which the row resides.

Write creation of each index type?

B-tree index:

```
CREATE <UNIQUE/NON UNIQUE> INDEX
```

```
<index_name>
```

```
ON <table_name> (<column_name>...)
```

```
TABLESPACE <tbs_name>;
```

Bitmap index:

```
CREATE BITMAP INDEX <index_name>
```

```
ON <table_name> (<column_name>...)
```

```
PCTFREE <integer>
```

```
TABLESPACE <tbs_name>
```

Function index:

```
CREATE INDEX <index_name>
```

```
ON <table_name>
```

```
[Function (<column_name>...)]
```

```
TABLESPACE <tbs_name>
```


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Reverse index:

```
CREATE INDEX <index-name>  
ON <table-name> (<column-name>  
TABLESPACE <tbs-name>  
REVERSE;
```